

Lecture 8, Feb 6 2026.

Example: Prove the identity $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$.

Sln. Let $f(x) = \arctan x + \operatorname{arccot} x$

$$\begin{aligned} f'(x) &= \frac{d}{dx}(\arctan x) + \frac{d}{dx}(\operatorname{arccot} x) \\ &= \frac{1}{1+x^2} + \left(\frac{-1}{1+x^2} \right) = 0 \quad \text{for all } x. \end{aligned}$$

then $f(x) = C$, where C is a constant.

$$C = f(1) = \arctan(1) + \operatorname{arccot}(1) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}.$$

then

$$\tan^{-1}(x) + \cot^{-1}(x) = \frac{\pi}{2}.$$

6.8. Indeterminate Forms and L'Hôpital's Rule.

Let consider $f(x) = \frac{\ln x}{x-1}$ if $x \rightarrow 1$ then $f(x) \rightarrow ?$

In general, if we have a limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where both $f(x) \rightarrow 0$, $g(x) \rightarrow 0$ as $x \rightarrow a$ then this limit may or may not exist and is called an indeterminate form of type $\frac{0}{0}$

$$\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x(x-1)}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{x}{x+1} = \frac{1}{2}.$$

$$\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{2x - 1}{2x} = \frac{2-1}{2} = \frac{1}{2}$$

Thm. (L'Hôpital's Rule). Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contain a .

assume $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$.

or that

$\lim_{x \rightarrow a} f(x) = \pm\infty$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$.

In other words, we have indeterminate forms of type $\frac{0}{0}$ / $\frac{\infty}{\infty}$.

Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

Example. Find $\lim_{x \rightarrow 1} \frac{\ln(x)}{x-1}$

Sln. Notice that $\lim_{x \rightarrow 1} \ln(x) = 0$ and $\lim_{x \rightarrow 1} (x-1) = 0$

and $\frac{d}{dx}(x-1) = 1 \neq 0$ By the L'Hôpital's rule

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\ln(x)}{x-1} &= \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)} \\ &= \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow 1} \frac{1}{x} = 1. \end{aligned}$$

Example. Calculate $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$.

Sln. Notice that $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow \infty} x^2 = \infty$.

We have $\frac{\infty}{\infty}$, and $\frac{d}{dx}(x^2) = 2x \neq 0$. By the L'Hôpital's

rule we have

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(e^x)}{\frac{d}{dx}(x^2)}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{2x}$$

$$\text{L'Hôpital rule} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(e^x)}{\frac{d}{dx}(2x)}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$$

Example. Find $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$.

Sln. By the L'Hôpital rule:

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\tan^2 x}{3x^2} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{2 \tan x \cdot \sec^2 x}{6x}$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\tan x \cdot \sec^2 x}{x}$$

$$= \frac{1}{3} \left(\lim_{x \rightarrow 0} \frac{\tan x}{x} \right) \cdot \underbrace{\lim_{x \rightarrow 0} \sec^2 x}_{1}$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin x}{x \cdot \cos x}$$

$$= \frac{1}{3} \underbrace{\lim_{x \rightarrow 0} \frac{\sin x}{x}}_1 \cdot \underbrace{\lim_{x \rightarrow 0} \frac{1}{\cos x}}_1 = \frac{1}{3}$$

L'H.

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sec^4 x + \cancel{\tan x (2 \sec x) \sec x \cdot \tan x}}{1} \rightarrow 0$$
$$= \frac{1}{3}$$

Example. Find $\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x}$.

Sln. $\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x} \stackrel{L'H}{=} \lim_{x \rightarrow \pi^-} \frac{\cos x}{\sin x} = -\infty$. *Wrong!*

Notice that $(1 - \cos x) \rightarrow 2$ as $x \rightarrow \pi^-$

$$\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x} = \frac{0}{2} = 0.$$